Date	15/07/2026
Team ID	xxxxxx
Project Name	Smart Lender (Loan Approval Prediction)
Maximum Marks	5 Marks

Proposed Solution Template

Project proposal and specifications report.

Project Overview

Objective: Develop a web application that evaluates loan eligibility parameters and predicts loan approval outcomes with high precision and low latency.

Scope: Data cleaning, categorical encoding, handling imbalance classes using SMOTE, training multiple classifiers (DT, RF, KNN, XGBoost), evaluating models, serializing the best model (Random Forest), and developing a Flask UI.

Problem Statement

Manual underwriting is slow, expensive, and subjective. Financial institutions require a reliable, automated tool to pre-screen applicants, filtering out high-risk profiles while approving creditworthy applicants based on historic loan performance records.

Resource Requirements

Resource Type	Description / Allocation	Specification
Hardware	Developer workstations	Intel Core i5/i7 Processor, 8GB+ RAM, 256GB SSD
Software	Development stack	Python 3.13, VS Code, Git, Flask web server
ML Libraries	Data science tools	Scikit-learn, joblib, pandas, numpy
Data Source	Historic loan application data	Dataset: train.csv (614 records)